SAKTHIVEL R

Artificial Intelligence & Machine Learning | Full-Stack Development
Chennai, Tamil Nadu | +91 9342212825 | sakthivel.aiml@gmail.com
LinkedIn: linkedin.com/in/r-sakthivelaiml | GitHub: github.com/sakthivel222 | Portfolio: portfolio-sakthivel222s-projects.vercel.app

PROFESSIONAL SUMMARY

Artificial Intelligence and Machine Learning engineering student with hands-on project experience in Python, generative AI, retrieval-augmented generation (RAG), and full-stack development. Familiar with building AI-enabled applications and web interfaces using React.js, FastAPI, LangChain, and vector databases. Seeking an internship opportunity to apply problem-solving skills and develop practical AI/ML solutions.

SKILLS

Programming: Python, Java, C++ **Databases:** MySQL, MongoDB
AI/ML: Machine Learning, NLP, Generative AI, LLMs, RAG, Prompt Engineering
Frameworks/Tools: React.js, FastAPI, Node.js, Express.js, Streamlit, LangChain, FAISS, ChromaDB, Git/GitHub
Core Skills: Data Structures and Algorithms, Problem Solving, Analytical Thinking, Communication, Time Management

PROJECTS

NAYAN-X – Predictive Road Hazard Navigation **AI/ML Concept**

- Proposed an AI-driven navigation approach to anticipate hidden or occluded road hazards and improve route safety.
- Combines computer-vision-based hazard perception, road-risk memory, and road-user intent prediction to support adaptive planning.

VEL Clinic – Physiotherapy Clinic Management App **Web Application**

- Built a clinic workflow application to organize patient details, appointments, treatment sessions, exercise information, calendar scheduling, and payment tracking.
- **Live Demo:** vel-clinic-appp-kypx.vercel.app

RAG AI Application **Generative AI**

- Built a retrieval-augmented generation workflow using LangChain, an LLM, prompt engineering, and vector search (FAISS/ChromaDB) for context-aware responses.

AI Resume Analyzer **Python, NLP, Streamlit**
Developed a resume-screening tool that analyzes resume content using NLP and machine-learning methods to support ATS-oriented review.

CampusOS AI **Generative AI, React.js, FastAPI**
Created an AI-powered campus assistant concept for academic support and campus-service queries.

E-Commerce Website **React.js, Node.js, Express.js, MongoDB**
Developed a responsive platform with authentication, product management, shopping cart, and payment integration.

EchoLife AI **Generative AI, RAG, Voice AI**
Developing a digital legacy platform concept using vector databases and cloud storage to preserve memories through intelligent conversations.

EDUCATION

B.E. in Artificial Intelligence & Machine Learning 2024–2028 (Pursuing)
Rajalakshmi Institute of Technology, Chennai, Tamil Nadu **CGPA: 8.07**

RELEVANT COURSEWORK

Data Structures and Algorithms, Database Management Systems, Object-Oriented Programming, Machine Learning, Natural Language Processing, Artificial Intelligence

CERTIFICATIONS

- Python for Data Science – IBM (2026)
- MongoDB Associate – MongoDB (2026)
- Machine Learning for Data Science Projects – IBM (2026)